

Accelerated observers in SRT

Y25 (2025) — English translation

Typically, special relativity (SRT) considers motion only relative to inertial reference systems. However, accelerated bodies can also be considered as observers. By connecting a reference frame to accelerated observers, one can obtain non-inertial analogs in SRT. This problem explores such constructions for one-dimensional motion along the Ox axis. Unless stated otherwise, positions are specified relative to an inertial laboratory frame with coordinates (t, x) .

Part A. Pseudo-relativistic accelerated observers (3.2 points)

Consider initially stationary point observers moving with constant acceleration in the non-relativistic sense. They exchange signals traveling at the speed of light c . The acceleration a of an observer depends on their initial coordinate d as:

$$a(d) = \frac{a_0}{1 + 2a_0d/c^2}, \quad d \geq 0.$$

A1	Find the maximum possible movement time $t_{\max}$ before the particle speed reaches c .	0.10
A2	Show that if for two observers $d_1 < d_2$ at $t = 0$, then for all $t < t_{\max}$, their coordinates satisfy $x_1(t) < x_2(t)$.	0.50
A3	Let observer 1 at time t_1 emit a signal received by observer 2 at t_2 . Show that for $d_2 > d_1$: $(t_1 - c/a_1)^2 = \varepsilon(t_2 - c/a_2)^2$, and for $d_2 < d_1$: $(t_1 + c/a_1)^2 = \varepsilon(t_2 + c/a_2)^2$. Express the constant ε in terms of a_1, a_2, d_1, d_2 .	1.00
A4	Observer 1 (at d_1) emits a signal at t_A , which reflects off observer 2 (at $d_2 > d_1$) at t_C and returns at t_B . The radar distance is $D_{rad} = c(t_B - t_A)/2$. Express D_{rad} in terms of d_1, d_2, a_0, c and show it is independent of t_A .	1.30
A5	For $d_2 = d_1 + \Delta d$ ($\Delta d \ll d_1$), express D_{rad} to first order in Δd in terms of d, a_0, c , and Δd .	0.30

Part B. Radar coordinates and Lorentz transformations (1.3 points)

An observer moves as $(t_O(\tau), x_O(\tau))$ where τ is proper time. To define coordinates (T, X) for an event at (t, x) , the observer sends a signal at proper time τ_1 that reaches the event, and receives the reflection at τ_2 . Then:

$$T = \frac{1}{2}(\tau_1 + \tau_2), \quad X = \pm \frac{c}{2}(\tau_2 - \tau_1)$$

where the sign depends on whether the event is to the right (+) or left (-) of the observer.

B1	If the observer moves with constant velocity v and $(t, x) = (0, 0)$ at $\tau = 0$, express x and t in terms of τ .	0.20
B2	For an event at (t, x) , find τ_1 and τ_2 for an observer moving at constant v .	0.70
B3	Show that (T, X) match the standard Lorentz transformed coordinates for a frame moving at velocity v .	0.40

Part C. Accelerated Observers (5.5 points)

Consider an observer with constant acceleration g in their instantaneous rest frame.

C1	Show that for suitable initial conditions: $x = A \cosh(\alpha\tau)$ and $t = B \sinh(\alpha\tau)$. Determine A, B , and α in terms of g and c .	0.80
C2	Coordinates (t, x) can be related to radar coordinates (T, X) as: $x = Af(X) \cosh(\alpha T)$ and $t = Bf(X) \sinh(\alpha T)$. Find the function $f(X)$ in terms of g and c . Sketch the region of space-time (x, t) covered by these coordinates as T, X vary from $-\infty$ to $+\infty$.	1.50
C3	Show the interval $ds^2 = c^2 dt^2 - dx^2$ can be written as $ds^2 = F(X, T)(c^2 dT^2 - dX^2)$. Find $F(X, T)$ in terms of g, c , and X .	0.50
C4	Bodies at $X = \text{const}$ are considered motionless in this system. Show they move with constant acceleration in their own rest frames and find this acceleration $a(X)$.	0.70
C5	A light ray is emitted from X_0 at T_0 in the $+x$ direction. How does X depend on T for this ray?	0.70
C6	An observer at $X_1 = \text{const}$ emits light of wavelength λ_1 . Determine the wavelength λ_2 measured by an observer at $X_2 = \text{const}$.	1.30

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